

NEO SUITE

ABAPConf 2026

Secure and automated ABAP Cloud
Full-Stack Development Process -
From Concept to Successful Delivery

<https://abapconf.org/abapconf2026/>

► Dr. Wolfgang Röckelein

Professional software developer since 1983, since 2005 in the ABAP world (and graduating in MIS plus achieving a PhD in between). I am passionate about converting customer requirements into working, delightful and maintainable software in my roles as consultant, software architect, and full-stack developer, since 2014 at AXIANS NEO (part of the VINCI group). I am active in the German speaking SAP user group both on the functional side (eg speaker of the working group on workforce scheduling) and on the technical side (eg as co-author of the ADT guideline). I live in the Munich area and from time to time I attend rock and goth events.

► Markus Bittner

Professional developer since September 2011. In 2012, I started working with SAP and SAPUI5 as a freelancer. In the beginning, I developed PoCs for SAPUI5 applications and have since experienced the entire evolution of the framework. Since 2016, I have been working at AXIANS NEO (part of the VINCI group), where I developed our Kapsel SDK applications using SAPUI5 as a UI Lead. Since 2019, I have taken on the role of UI Lead for NEO Schedule. My hobbies are gaming, brewing beer, and attending medieval festivals.

abap
conf2021

11:35



Agile ABAP Produkt-Entwicklung mit Clean Code,
Unit Tests und ATC

Speaker [Dr. Wolfgang Röcklein](#)

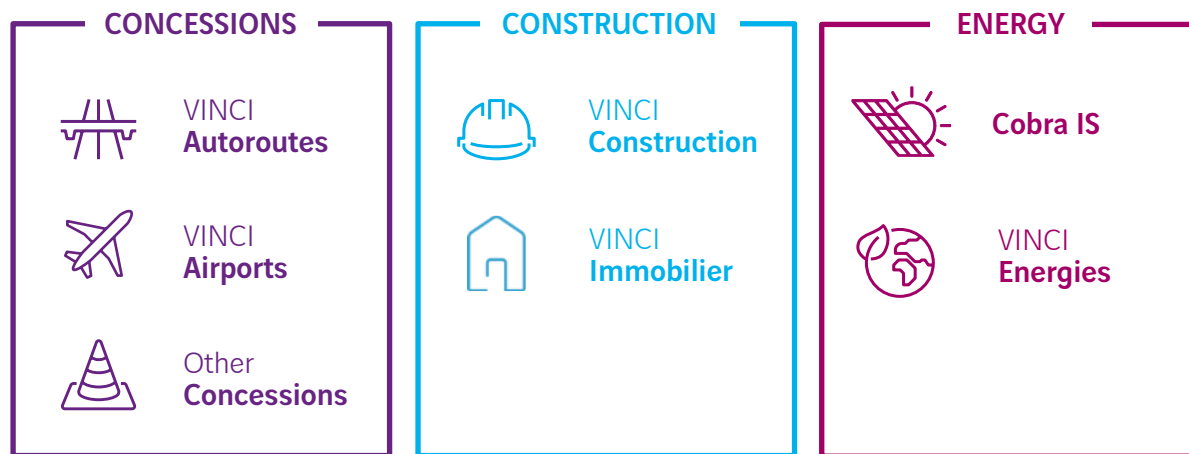
Duration 45 Min

Language Deutsch

 [Download slides](#)

VINCI AT A GLANCE

axians

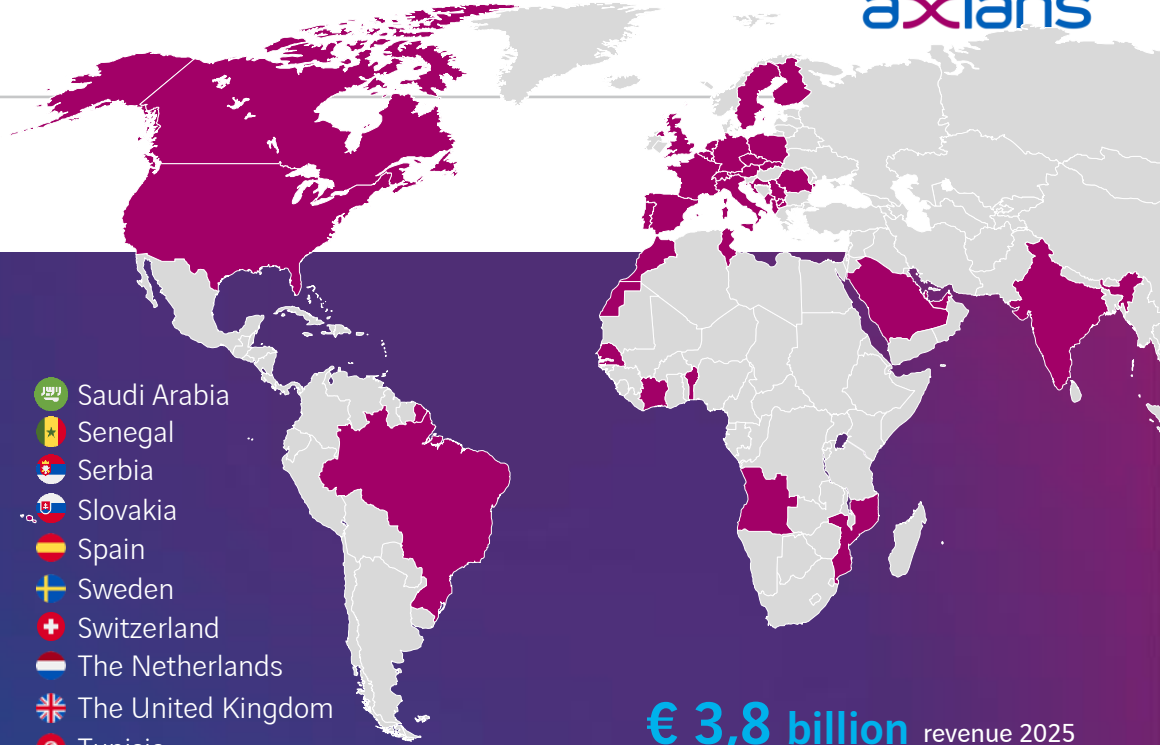


€ 74,6 billion
revenue 2025

>120
Countries

294 k
Employees

► We know no boundaries



- | | | |
|--|---|--|
|  Albania |  Italy |  Saudi Arabia |
|  Angola |  Ivory Coast |  Senegal |
|  Austria |  Kosovo |  Serbia |
|  Belgium |  Luxembourg |  Slovakia |
|  Benin |  Monaco |  Spain |
|  Brazil |  Montenegro |  Sweden |
|  Canada |  Morocco |  Switzerland |
|  Croatia |  Mozambique |  The Netherlands |
|  Czech Republic |  North Macedonia |  The United Kingdom |
|  Finland |  Poland |  Tunisia |
|  France/
French overseas |  Portugal |  United Arab Emirates |
|  Germany |  Qatar |  USA |
|  India |  Romania | |

€ 3,8 billion revenue 2025

17K Employees

38 Countries

NEO Service-Portal



Information & communication, customer loyalty around the clock

NEO Schedule



The right resources, in the right place, at the right time

NEO Mobile



The digital assistant in everyday technical work

NEO Expert Service



Virtual on-site support for technicians and customers

NEO CBM (IoT)



IoT-based real-time analytics for a dynamic maintenance approach

Field-Service-Management

NEO Insight

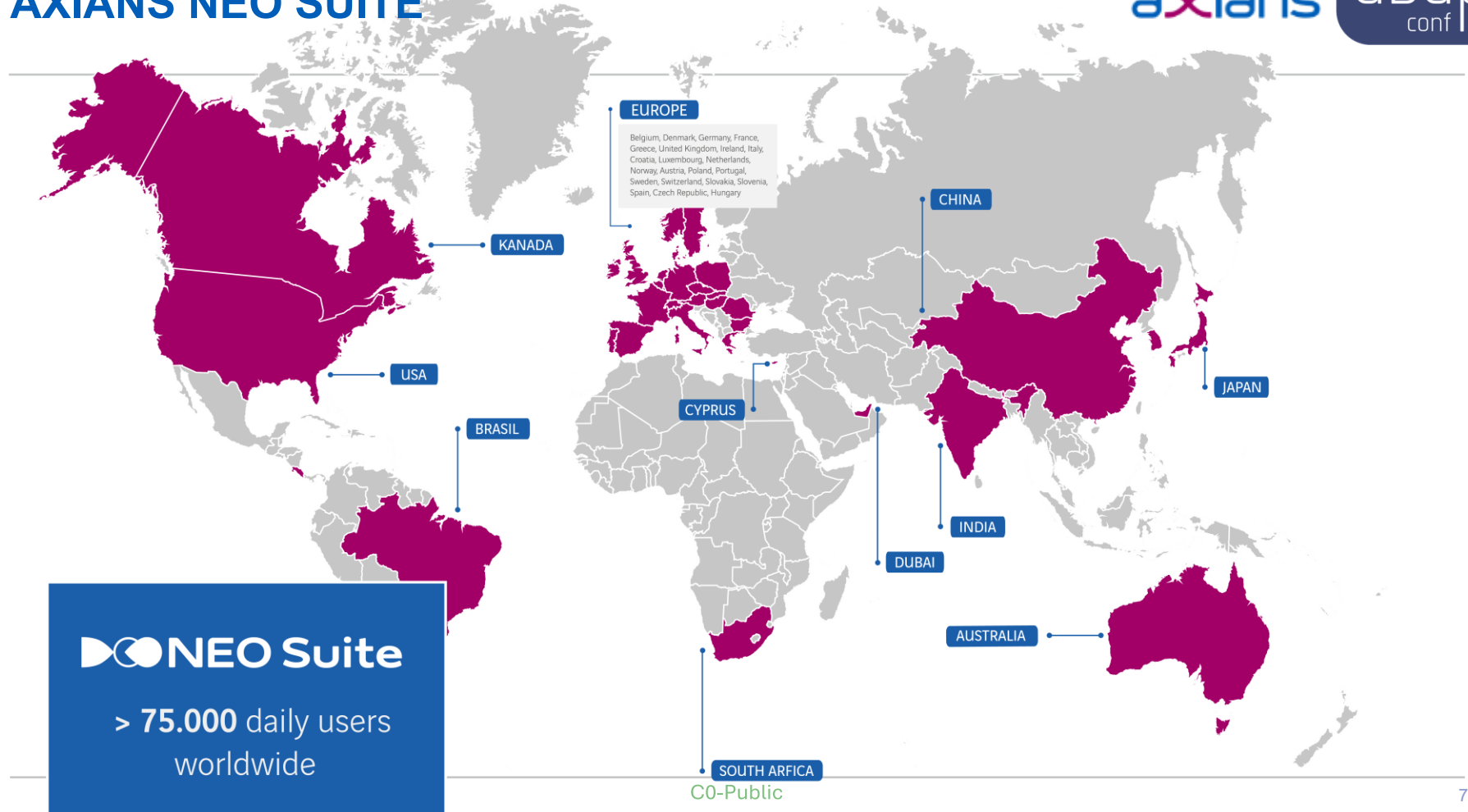


Key performance indicator assisted decision-making in maintenance and customer service

AXIANS NEO SUITE

axians

abap
conf

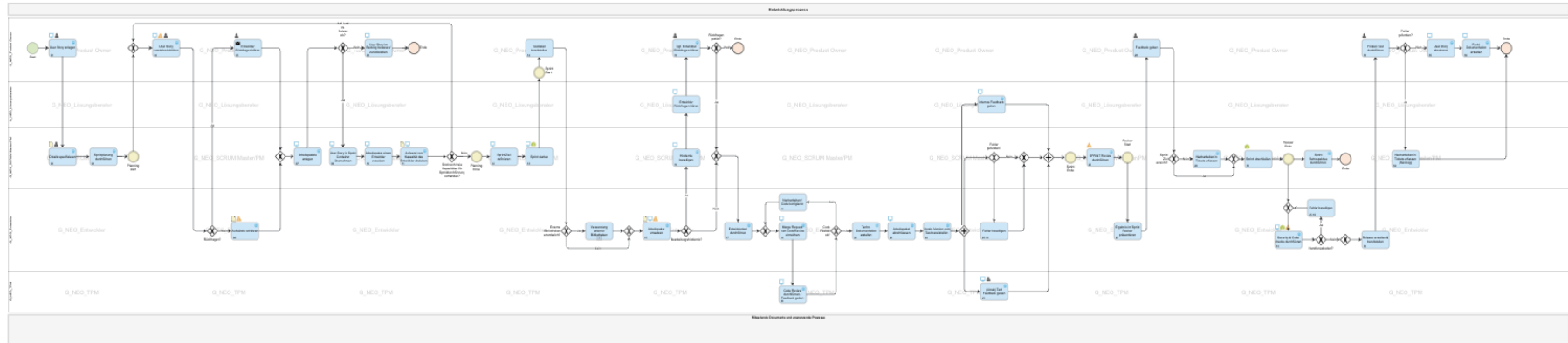


► Certification of the Business Unit since 2025

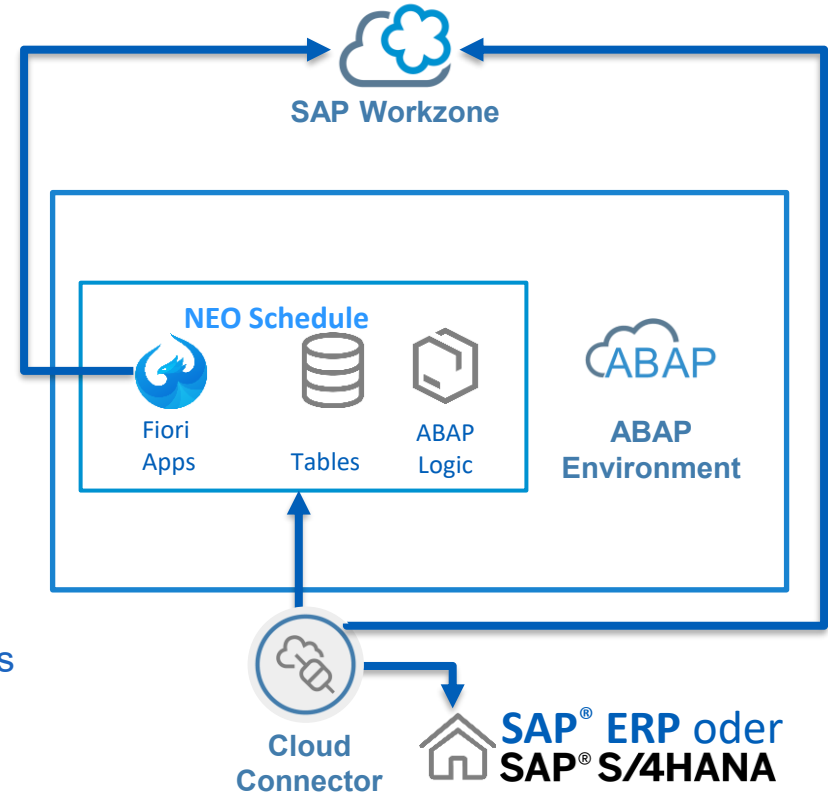
- Group Certification in the Axians Group
- United We Stand
- and United We Would Fall – But: United We Never Shall Fall



► Living a secure development process since the initial LoC of NEO Schedule



- ▶ Core Unit as Full Stack Solution available for On-Stack and Side-by-Side Deployment
 - Side-by-Side Deployment utilizing Customer owned BTP ABAP Environment System with Add-On Delivery
 - Fiori Apps in ABAP Env available for BTP Workzone Federation
 - In BTP Workzone Fiori Apps from S/4 System can also be federated
- ▶ Modular Integration Units deployed in Source Systems (from EHP7 up to Current S/4 Release)
- ▶ Pure Remote Integration Units without Solution Artefacts in Source System Upcoming



► Backend

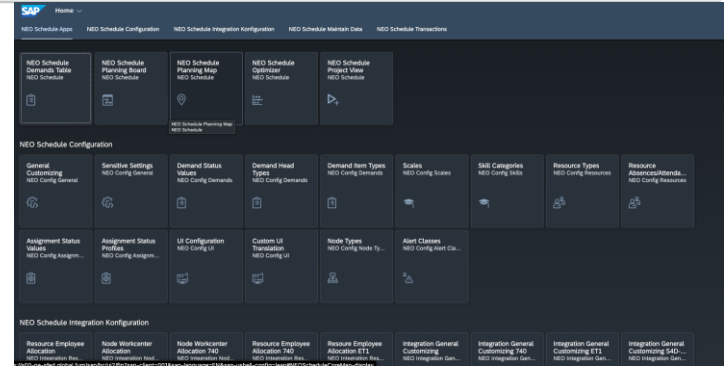
- Standard RAP Developer Extensibility esp. for Field Extensibility
- Core DAO Classes Extensible with Before/After CUD Extension Points
- Almost all Classes extensible (Non-Final, Factory Pattern, Methods public/protected)
- Configuration in Business Configurations

► Frontend

- UI5 Standard Extensibility
 - App Extension, UI Adaptation, Metadata driven (Fiori Elements + Building Blocks)
- Custom Extensibility Framework
 - Release Stable Interfaces
 - Without Redeployment and load Extensions at Runtime
 - Follow the UI5 MVC Standard
 - Custom NEO XML Fragments (V)
 - ES6 Classes Base+NonBase (C)
- Frontend Configuration via entries in Backend Business Configurations

NEO SCHEDULE FRONTEND (1)

- Fiori Element Apps (32)
 - Configuration (16 Apps)
 - Obsolete in ABAP Environment
 - Custom Business Configurations
 - Data management (10 Apps)
 - Integration (6 Apps)



SAP Resources

Standard

Search

Description: *Markus* 1 more

Time Zone: CET (Central Europe)

Resource Typegroup: Interne Mitarbeiter

Email Address:

Phone Number:

User of Resource:

Integration Id:

Number:

Validity Begin:

Validity End:

Sort Number:

Go Adapt Filters (1)

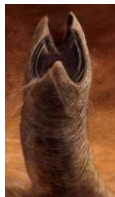
Description	Time Zone	Resource Typegroup	Phone Number	User of Resource	Email Address	Validity End
Markus Bitner Sort Number: 0	CET (Central Europe)	Interne Mitarbeiter				12/31/2065, 23:59:59
Wolfgang Röcklein Sort Number: 0	CET (Central Europe)	WORO_EN				12/31/2065, 23:59:59

► Freestyle Apps (5 + 2 Embedded apps)

- Building Blocks aka Metadata Driven (Filterbar, Tables, etc)
- Using UI5 Controls only
- Drag and Drop between Apps
- Push (WebSockets/SharedWorker)
 - App Automated Background Refreshes
 - App to App Communication
- Custom Extension Framework



- ▶ Shai-Hulud
 - September 2025
- ▶ Shai-Hulud 2.0
 - November 2025
- ▶ Trivy compromization
 - March 2026
- ▶ Axios compromization
 - March 2026
- ▶ Mini Shai-Hulud
 - April 2026
 - @cap-js/sqlite@2.2.2
 - @cap-js/postgres@2.2.2
 - @cap-js/db-service@2.10.1
 - mbt@1.2.48
- ▶ Targeting API Keys (compromising Dev Machines)



HUNDERTE NPM-PAKETE BETROFFEN

Ein Wurm frisst sich durch das Javascript-Ökosystem

Im Javascript-Ökosystem breitet sich eine Malware mit wurmartigem Verhalten aus. Fast 500 NPM-Paketversionen sollen bereits kompromittiert sein.

17. September 2025 um 10:14 Uhr / Marc Stöckel

Trivy Compromised: Everything You Need to Know about the Latest Supply Chain Attack

On March 19, 2026, threat actors injected credential-stealing malware into Actions. Learn how "TeamPCP" executed this breach and how to audit your

MAR 31, 2026

One of the most popular JavaScript packages on earth Axios has been compromised

The Axios NPM package has been compromised and the maintainer of the project has been locked out of their account. This will go down in history as one of the most successful software supply chain attacks ever

Mini Shai Hulud and SAP Compromise

MALWARE

SafeDep Team • Apr 29, 2026 • 12 min read

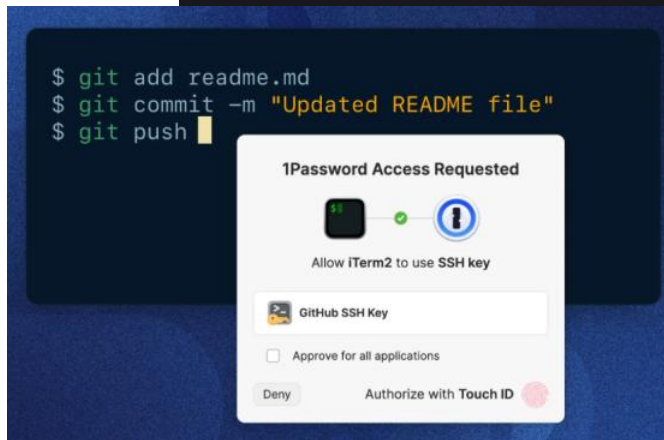
npm tackles its riskiest security issues

With npm v12, GitHub closes a central attack vector: installation scripts from dependencies will only run after explicit approval from July 2026.

- ▶ BoM in all Development Stages
 - Automated Security Checks
 - Direct & Indirect Dependencies
 - Cancel Pipelines on Security Issues
- ▶ Secrets Management
 - No plain text secrets
 - No Secrets on File System
 - `op://<vault>/<item>/<field>` (1Password cli)
- ▶ Dedicated GitLab Runner
 - Minimal user rights
 - Secure VMs
- ▶ Only Automated Deployment

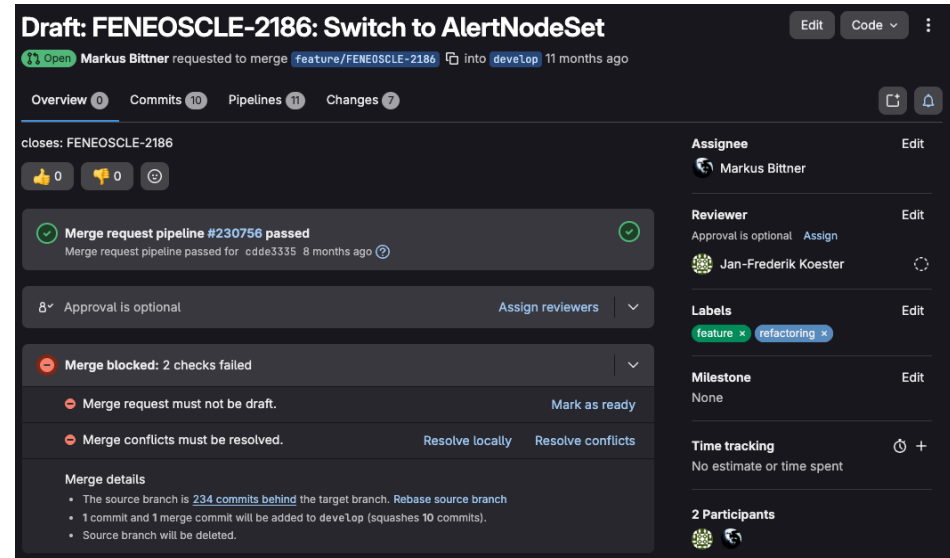
Transport Request "<concealed by 1Password>" has been

✓ Passed	#1059402	util	4c752fd9	2 days ago
✗ Failed	#1059396	util	5cd52550	2 days ago
✗ Failed	#1059370	util	0137aa9c	2 days ago
✓ Passed	#1052550	NEO Schedule Planning Board	7800d433	6 days ago
✓ Passed	#1052549	NEO Schedule Planning Board	7800d433	6 days ago
			00d433	6 days ago
			2c29a0	6 days ago



<https://1password.com/developer-security>

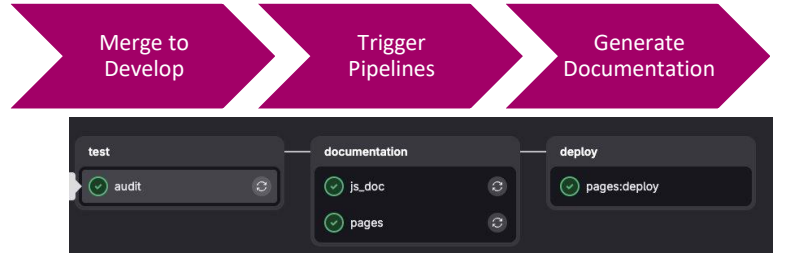
- ▶ Axians Internal GitLab Instance
- ▶ GitLab Merge Requests
 - Working with Feature Branches
 - Four Eyes Principle
- ▶ Review
 - Security Checks (By Reviewer)
 - Business Logic + Tests
 - Automate with AI in future (maybe)



The screenshot shows a GitLab Merge Request (MR) interface. At the top, it says "Draft: FENEOSCLE-2186: Switch to AlertNodeSet". Below this, it indicates that "Markus Bittner requested to merge feature/FENEOSCLE-2186 into develop 11 months ago". The interface has tabs for Overview (0), Commits (10), Pipelines (11), and Changes (7). The Overview tab is active, showing a list of changes and a section for "Merge request pipeline #230756 passed". Below this, there are sections for "Approval is optional" and "Merge blocked: 2 checks failed". The "Merge blocked" section lists two checks: "Merge request must not be draft." and "Merge conflicts must be resolved.". The "Merge details" section at the bottom provides information about the source branch, including the number of commits behind the target branch and the number of commits to be added to the target branch upon merge.

CI/CD

- Security Checks
 - npm audit (Cancel @ High)
- Generate Documentation
 - jsdoc
- Deploy to ABAP Environment
 - @sap/ux-ui5-tooling
 - Mixin of Translations from SAP BTP Translation Hub



```
markusbittner@Mac core-gantt % op run npm audit
# npm audit report

form-data 4.0.0 - 4.0.5
Severity: high
form-data: CRLF injection in form-data via unescaped multipart field names and filenames - https://github.com/advisories/GHSA-hmw2-7cc7-3qxx
fix available via `npm audit fix`
node_modules/form-data
```

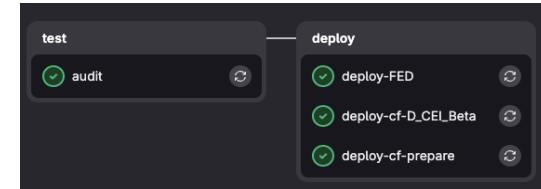
Failed Jobs 1

#351262 Merge branch 'develop' into feature/FEN... 00:00:04 2 hours ago

test: failed stage: test

Failed jobs

audit



- ▶ 219 Global Classes
- ▶ 45 Interfaces
- ▶ 101 Domains
- ▶ 235 Data Elements
- ▶ 156 Database Tables (incl. some draft tables...)
- ▶ 597 CDS Views (with RAP you end up with many CDS Views..)
- ▶ 77 Metadata Extensions
- ▶ 77 Behavior Definitions
- ▶ 46 Function Modules (for calls from source systems)

- ▶ Main Dev Artefacts: DDIC Objects, RAP BDEF, CDS Views, ABAP Classes
 - Core CDS Views and Core ABAP Classes have been ABAP Cloud Conform right from the start
 - Ensured by
 - ATC Checks
 - abapgit based Code Transfer between Onpremise and ABAP Environment Systems
- ▶ Unit Tests
 - Mocking and Test-Seams for external dependencies, Test Doubles fully Available in ABAP Env!
 - Coverage Measurement and Metering
 - Acceptance Problematic / Advantages Big
- ▶ Separation of Concerns through Small Classes
 - eg pre/post CUD extension points used in the solution itself
- ▶ Agile Automated Translation via SAP BTP Translation Hub
- ▶ Fine Granular Authorization Concept exposed via Role-Based Access Control and Attribute-Based Access Control -> Security also for the Deployed Solution

► Guidelines

- Massively Reduced Hungarian Notation – only parameter and main artefact (CL/IF/C/I) prefixes
- Glossary
- ABAPdoc

► Uniform Naming/Formatting Convention for Easy Code Reviews

► Code Reviews

- Ticket/Transport/Commit Based
- Human Understandable Code esp. Identifiers
 - Martin Fowler (1999): Any fool can write code that a computer can understand.

Good programmers write code that humans can understand.

- Open Discussion



► Semi-Automated External Documentation through ABAPdoc Export

- Easy code review of some documentation aspects esp. regarding customization entries
- ABAPdoc exported and published for every release

►  /NEO/CL_SKD_CALC_TRAVEL ►

```

1 | <p class="shorttext synchronized" lang="en">NEO Schedule: Travel Calculation Class</p>
2 | Implements {@link /neo/if_skd_ass_dao_ext.METH:insert_pre}, {@link /neo/if_skd_ass_dao_ext.METH:update_pre},
3 | {@link /neo/if_skd_ass_dao_ext.METH:insert_post}, {@link /neo/if_skd_ass_dao_ext.METH:update_post} and {@link /neo/if_
4 | <br>
5 | Also provides the Implementations for Align and Reorder Assignments Actions.
6 | <br>
7 | All implemented {@link /neo/if_skd_ass_dao_ext} methods need to be assigned in order to get consistent results!
8 | <br>
9 | Needs a {@link /neo/if_api_geo_functions} implementation which gives identical results for identical inputs at least d
10 | This will stop a cascade since only really changed assignments are updated.
11 | <br>

```

Class /NEO/CL_SKD_CALC_TRAVEL

```

12 | Logging ID: calc_travel cf {@link /neo/cl_skd_flat_config}/{@link /neo/cl_api
13 |
14 | <h3>The following parameters are supported (via {@link /neo/cl_skd_calc_travel
15 | <ul><li><strong>calc_travel_max_secs_between_ass4next_prev</strong> Maximum d
16 | <li><strong>calc_travel_default_distance_uom</strong> Default UoM for Travel.
17 | <li><strong>align_ass_max_search_for_begin_sec</strong> Maximum duration in s
18 | <li><strong>align_ass_longest_dur_of_range_sec</strong> Maximum duration in s
19 | <li><strong>align_ass_max_dur_abs_to_skip_sec</strong> Maximum duration of sm
20 | <li><strong>align_max_home_travel_absence_sec</strong> Maximum duration of as
21 | <li><strong>align_max_search_for_end_sec</strong> Maximum duration to search
22 | <li><strong>align_time_diff_for_stapling</strong> When assignment are left ov
23 | <h3>Related parameters (used in CDS views, values need thus to be in Short va
24 | <ul><li><strong>geo_url_prefix</strong> Prefix to use for construction of ma
25 | <li><strong>geo_url_infix</strong> Infix between Latitude and Longitude to us
26 | Alternative combination for the last two parameters would be
27 | <ul><li><em>https://www.google.com/maps/search/?api=1&query=</em> and <em>
28 | <li><em>https://www.google.com/maps/&#64;?api=1&map_action=map&center=</
29 | <li><em>geo:</em> and <em>.</em> Attention: does not work in windows</li>
30 |
31 | Uses current Geo Implementation from {@link /neo/cl_api_geo_base}
32 | CLASS /neo/cl_skd_calc_travel DEFINITION PUBLIC INHERITING FROM /neo/cl_skd_all_
33 |

```

Documentation

NEO Schedule: Travel Calculation Class

Implements *insert_pre*, *update_pre*, *insert_post* and *delete_post*
Also provides the Implementations for Align and Reorder Assignments Actions.

All implemented */neo/if_skd_ass_dao_ext* methods need to be assigned in order to get consistent results!

Needs a */neo/if_api_geo_functions* implementation which gives identical results for identical inputs at least during a short timeframe (ie during one DAO transaction).
Logging ID: calc_travel cf */neo/cl_skd_flat_config*/*/neo/cl_api_logger_base*

The following parameters are supported (via *constructor*)

- *calc_travel_max_secs_between_ass4next_prev* Maximum duration for assignment travel in seconds. For greater gaps travel to/from resource
- *calc_travel_default_distance_uom* Default UoM for Travel. Default is KM
- *align_ass_max_search_for_begin_sec* Maximum duration in seconds for search for begin for align assignments. Default is 36h.
- *align_ass_longest_dur_of_range_sec* Maximum duration in seconds for search for end for align/reorder assignments. Default is 24h.
- *align_ass_max_dur_abs_to_skip_sec* Maximum duration of small absences to skip. Default is 15min.
- *align_max_home_travel_absence_sec* Maximum duration of assignment+home travel in absence. When more, this and following assignments are "stapled"
- *align_max_search_for_end_sec* Maximum duration to search for shift end (backward/forward) for move to next shift function. Default is 2h
- *align_time_diff_for_stapling* When assignment are left over after align, they are "stapled" after shift end, preserving order by using this amount in seconds be

Related parameters (used in CDS views, values need thus to be in Short Value Field)

- *geo_url_prefix* Prefix to use for construction of map URLs. Default is *bingmaps/?cp=*
- *geo_url_infix* Infix between Latitude and Longitude to use for construction of map URLs. Default is *~*

Alternative combination for the last two parameters would be

- *'https://www.google.com/maps/search/?api=1&query='* and *'*
- *'https://www.google.com/maps/@?api=1&map_action=map¢er='* and *'*
- *'geo'* and *'* Attention: does not work in windows

Uses current Geo Implementation from */neo/cl_api_geo_base*

► ATC Checks at various Stages

- During Coding
- During Transport Release
- During abapgit Commit/Push (onpremise)
- During Release Build in Build System
- Includes Unit Tests

► Different ATC Check Variants used

- SLIN_SEC (aka CVA)
 - No Exemptions! Zero Findings Policy!
- DEFAULT (and our ZDEFAULT)
- SAP_CP_READINESS_BTPLATEST
- ABAP_CLOUD_DEVELOPMENT_PARTNER
- No Errors Policy! Only Minimal Warnings Policy!

► Utilizing the ABAP Environment as Central Check System for OnPrem Dev Systems since Availability (2311)

- Previously we Used Onpremise Central Check System (S/4 System)

ATC Check Variant: ABAP_CLOUD_DEVELOPMENT_PARTNER

ATC Checks

Select the ATC checks you want to include in the ATC check variant.

- ☐ ABAP Cloud Readiness
- ☒ ABAP Dictionary Checks
- ☒ ABAP RESTful Programming Model
- ☒ API Release
- ☐ ATC Check Category for Proxy Objects
- ☐ BAPI Konsistenz-Check
- ☐ Category for tests validating RAP BO contracts
- ☐ checks for KTD
- ☐ Clean Core
- ☒ Core Data Services (CDS)
- ☒ Dynamic Code Checks
 - ☒ ABAP Unit Check <remote-enabled>
- ☐ Enterprise Search Modeling
- ☐ General Checks
- ☐ Identity and Access Management
- ☒ ILM (Information Lifecycle Management)
- ☒ Performance
- ☐ Programming Conventions
- ☒ Robust programming
- ☐ S/4HANA Custom Code Checks
- ☐ S/4HANA: Idoc Check
- ☐ Search functions
- ☒ Security
- ☒ Software Component Relations
- ☐ STC Technical Configurations
- ☒ Syntax and activation
- ☒ User Interfaces

► NSC_EN ► SLIN_SEC ► Last Check Run

type filter text

Description

 Findings: None

The [ABAP Unit](#) adapter for (transaction SCI) Code Inspector executes unit tests and converts Code Inspector result format.

Notes

ABAP Unit tests are not a static analysis. Unit tests are code fragments that invoke the domain purposes. Alerts such as those triggered by calling methods of the class [cl_Abap_Unit_Assess](#) containing the tests.

SAP Display ATC Settings

✓

Code Inspector

Global Check Variant

Handling of Pragmas/Pseudo Comment

Reference Check System

Default Check Options

☐ Analyze Enhanced Objects

Allow additional features in ABAP Development Tools?

☐ No

☒ Yes

RFC Destination

► Controlled Build through automated build pipeline offered by SAP

- Includes ATC and other Checks
- Controlled RTC (Release to Customers)
 - after Brief Manual Tests in our Test System – Human in the Loop

Build Product Version ▾

Build Status for Patch Delivery 2.1.12 of Product /NEO/SKD_APIMGT Refresh

Init Initial Checks Prepare Sys... Clone Repo... **ATC** Build Integration ... Publish

[2026-06-17T00:27:53.047Z] info abapEnvironmentRunATCCheck - ATC run completed successfully. If there are any results from the respective run they will be listed in the logs above as well as being saved in the output .xml file

[2026-06-17T00:27:53.047Z] info abapEnvironmentRunATCCheck - SUCCESS

[2026-06-17T00:27:53.047Z] info abapEnvironmentRunATCCheck - Step telemetry data:{"StepStartTime":"2026-06-17 00:00:30.975873388 +0000 UTC","PipelineURLHash":"50f013532a9770a2c2cfdc38b7581dd01df69b70","BuildURLHash":"50f013532a9770a2c2cfdc38b7581dd01df69b70","StageName":"ATC","StepName":"abapEnvironmentRunATCCheck","ErrorCode":"0","StepDuration":"1638798","ErrorCategory":"undefined","CorrelationID":"n/a","PiperCommitHash":"8364e22e22aa","ActionVersion":"n/a","TemplateVersion":"n/a"}

[2026-06-17T00:27:53.650Z] Archiving artifacts

- ▶ Our Colleagues Willingness and our Managements Support
 - You have to Convince and not to Command
- ▶ Secure Development is a Mind Set – Old Habits Die Hard
- ▶ CVA/SLIN_SEC is an enormous help but not enough – coding must be done with security in mind
- ▶ Keep your eyes open for security issues in your environment – Security breaches are mostly silent

- ▶ All would not have been possible without (random and incomplete list!):
 - Open Source Efforts like abapgit
 - and Adoption of abapgit by SAP for ABAP Environment
 - ABAPdoc
 - SAP UI5 and OpenUI5 and Fiori Style Guides for Delightful and Uniform UIs
 - transferring lots of the good old Dynpro Ideas to the Web Future
 - SAP BTP Services like Translation Hub, Automated Builds (Project Piper)
 - Willingness of SAP to modernize ABAP (hey, we even get native Boolean now!)
 - CVA/SLIN_SEC on ABAP Environment for FREE esp as Central Check System
 - Open Discussions through SAP Community (Online and Local – Hi to all Stammtisch Sisters and Brothers!) and DSAG Groups

► Thank you

- To the Customers of our products for trusting us
- For Organizing abapconf
- For Participating as Speakers and Attendees at abapconf
- For the Keynote Opportunity – We are honored!

► Dr. Wolfgang Röckelein

- wolfgang.roeckelein@axians.de
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- <https://www.linkedin.com/in/wolfgang-roeckelein/>
- <https://profile.sap.com/u/WRoeckelein>
- <https://dsagnet.de/benutzer/a76aada4-7cf1-4720-80c7-18a6c37a9a73>

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